Installation and Operation Manual

GreenPower

GN27-40.5kV

Indoor Disconnector





About Us»

GreenPower promotes environmental awareness, and aims to create unprecedented happiness and wealth for our investors, employees, clients and partners. By focusing in the mid-to-high voltage and low voltage eld, and professionally working on R&D, manufacturing, marketing and service of high-end green intelligent switchgears, equipments and products.GreenPower sets to become a well-respected global company in the power industry.

GreenPower, is jointly established by a number of state-owned excellent industrial electrical professional manufacturers, we committed to meet the procurement needs of clients. It is a professional procurement service provider with entity of industrial electrical.

GreenPower provides ONE-STOP solutions service for all customers in the International area. It is located in the time-honored electrical industrial city, a collection of worldclass brand of industrial electrical products,excellent brand of domestic electrical products as the basis.

Innovative business philosophy, strong professional supply team, expert technical guidance, advanced Information network management platform, fast logistics, which makes GreenPower as your procurement expert by your side.

GreenPower adheres to its own brand and multi-brand integration, marketing dierentiated development strategy. The products involving low voltage, medium voltage and high voltage transmission and distribution products and industrial automation products.

It covering all aspects of distribution, logistics, warehousing, professional and technical engineering services, systems integration and complete sets of manufacturing. Stable business foundation for cooperation with customers and suppliers, and has established a good reputation to maintain its leading position in the market competition.

Human quests for transcendence, with transcendence we make progress, when the pursuit of unlimited become a belief, our dream began to realize.

Hard-working, dedicated GreenPower people, will be adhering to its past glory, beginners mind, let go sailing, to face the fierce competition in the future, and always help customers make the best choice, and strive to become the best and most reliable procurement service provider in the eld of industrial electrical.

01 Product overview

The GN27-40.5kV indoor high-voltage disconnectors (hereafter "disconnectors"), designed for manual operation with the CS6-1 operating mechanism, are intended for 40.5kV, 50/60Hz AC systems to open and close circuits under no-load conditions, and are designed and type-tested in accordance with the IEC 62271-103 standard.

02 Service conditions

- Altitude: 1000 m to 3000 m.
- Ambient air temperature: +40℃ max, -25℃ min; daily average ≤ +35℃ .
- Relative humidity (ambient): daily average ≤ 95%, monthly average ≤ 90%.
- Seismic intensity: not exceeding degree 8.
- Site conditions: free from fire/explosion hazard, severe pollution, chemical corrosion, and severe vibration.

Note: If operating conditions exceed the above limits, they shall be determined by consultation between the user and the manufacturer.

03 Main technical parameters

Model	Rated voltage (kV)	Rated current (A)	4s Thermal withstand (r.m.s.) (kA)	Dynamic withstand (peak) (kA)	Rated lightning impulse withstand voltage (kV)
GN27-40.5(C)(D)/630A	40.5	630	20	50	200
GN27-40.5(C)(D)/1250A		1250	31.5	80	
GN27-40.5(C)(D)/1600A		1600	31.5	80	
GN27-40.5(C)(D)/2000A		2000	40	100	

04 Structural features

The disconnector adopts a three-phase, common-base knife-blade structure, capable of opening/closing under voltage without load. The frame and supports for the current-carrying parts are welded from the main shaft, bent plates and angle steel. Six insulators are mounted on the base (two per phase), and three operating insulators per phase drive the blades for opening and closing. The current path consists of a fixed contact (female contact) and a contact base. The fixed contact is fitted with 6/14/16 pairs of spring-loaded contact fingers and allows vertical compliance to ensure reliable engagement. Each phase has two blades mounted on the contact base and spring-loaded.

Opening: The operating crank turns the shaft, the operating insulators push upward, separating the blades from the fixed contacts. The blades rotate about the contact base while the fixed contacts are driven upward to the open position.

Closing: The crank turns the shaft in the opposite direction; the operating insulators pull the blades downward. After the blades meet the fixed contacts, the fixed contacts are rotated with the blades until they reach the fully closed position together.

05 Outline and mounting dimensions

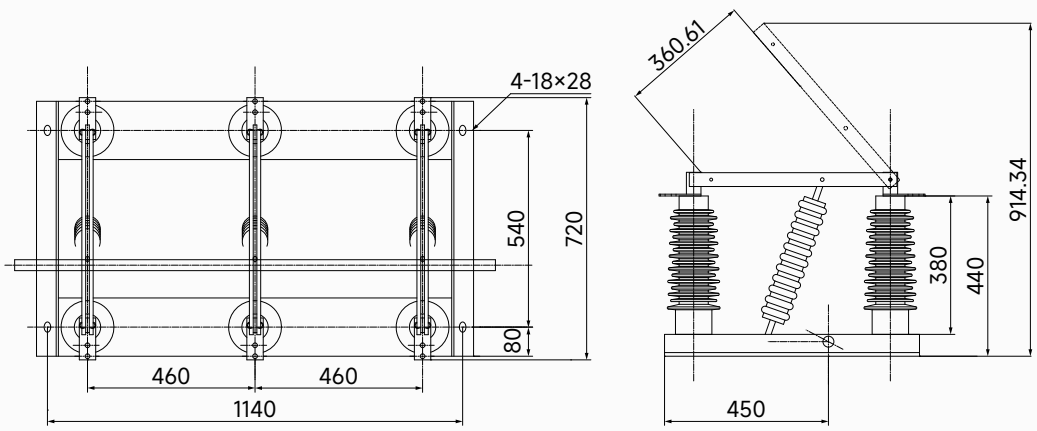


Figure 1 GN27-40.5/630A-460

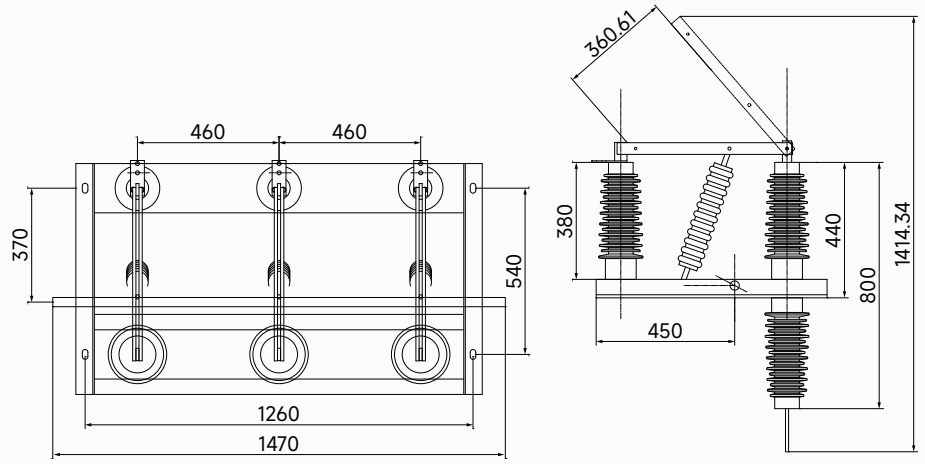


Figure 2 GN27-40.5C1/630A-460

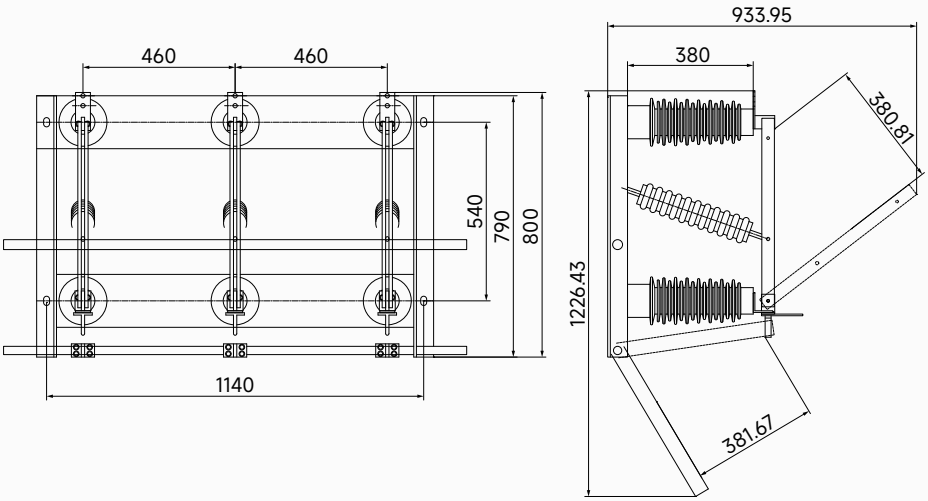


Figure 3 GN27-40.5D1/630A-460

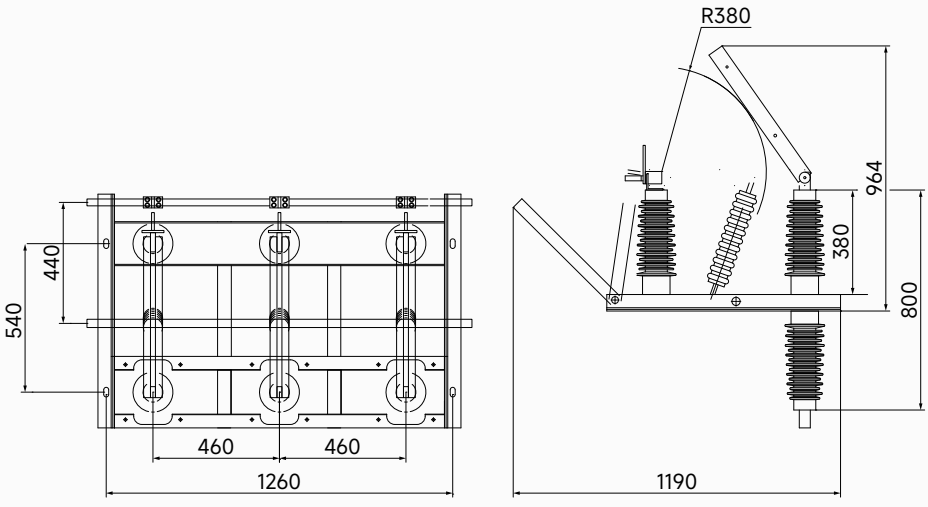


Figure 4 GN27-40.5C1D2/1600A-460

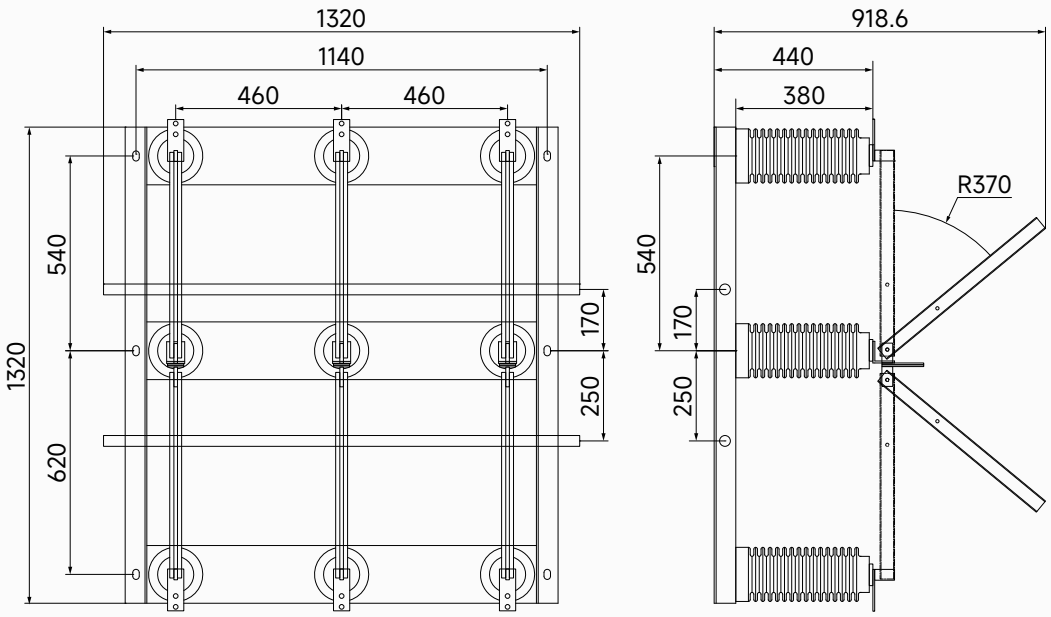


Figure 5 GN27-40.5ST/630A-460

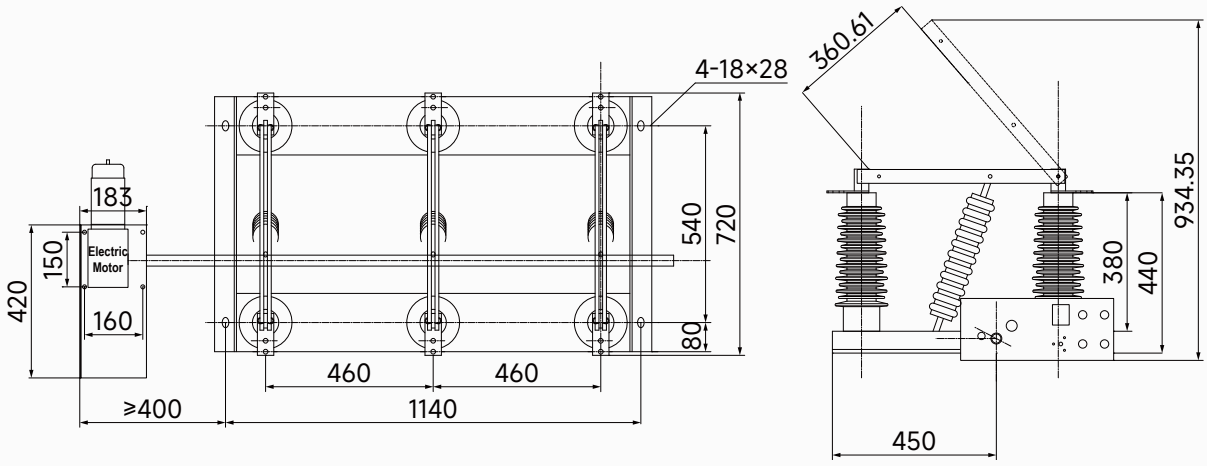


Figure 6 GN27-40.5MJ/630A-460

06 Ordering information

- When ordering, please specify the model/type, rated voltage, rated current, and other technical data.
- For special requirements, please consult the manufacturer for a tailored solution.

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Professional Power Equipment Provider

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